Ivan Melnikov

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Technical

Kotlin, Jetpack Compose, Coroutines, Android SDK, PyTorch, TypeScript, Svelte, Node.js, BLE

Languages

Russian, English, Italian, French, Latin

Experience

Software Engineer, Google Photos | Google LLC **August 2025 - ongoing**

- Designing and building agentic AI infrastructure for Google Photos, architecting dependency injection and lifecycle-aware scoping solutions across Binder, Dagger/Hilt, Fragment, Activity, and View boundaries to support cross-team adoption. Ask Photos is the first client of this framework.
- Modernized the Glide image library's open-source publishing pipeline — migrated build scripts from Groovy to Kotlin DSL and updated Maven publishing via GitHub Actions, unblocking the first release in over two years.

Software Engineer, Automotive Assistant | Google LLC **January 2021 - August 2025**

- Prototyped Gemini Live integration in Automotive Assistant using Jetpack Compose, demonstrated to press and OEMs at Google I/O. Drove adoption of Compose in a legacy codebase as a greenfield proving ground.
- Championed Kotlin adoption since joining in 2021, driving migration from legacy patterns and mentoring teammates on idiomatic patterns.
- Conceptualized, architected, and maintained the full DevOps pipeline — Play Store signing, automated testing, and artifact storage — impacting thousands of testers across multiple countries and OEMs.
- Managed initial release cycles coordinating with global teams, growing the fleet from 0 to 3 million vehicles.
- Implemented Acoustic Echo Cancellation for the Hotword detection system, synchronizing audio across multiple microphones to cancel background noise.
- Designed on-device Hotword diagnostics including custom logging, heartbeat service, and debug data providers. Developed end-to-end module for parsing diagnostic data.

Software Engineer, Bose Corporation **2017 - 2021**

- Modeled and implemented Bose Music features used by hundreds of thousands of users.
- Engineered CI/CD pipeline for testers and customers; decreased build times from 30 to 15 minutes by optimizing Gradle logic.
- Prototyped protocol for JSON message passing over BLE for media device control.
- Led weekly huddles with 50+ developers covering code reviews, architecture, and best practices.
- Rewrote core screens in SoundTouch app to optimize scalability across multiple resolutions and millions of customers.

Education

Johns Hopkins University **2023 - 2025**

- MSc Artificial Intelligence

Worcester Polytechnic Institute **2013 - 2017**

- BSc Computer Science
- BSc Nuclear Engineering

Projects

Sleep stage classification from wearable PPG and accelerometer data **August 2025**

- Developing a quantized 1D CNN for sleep stage classification using wearable PPG and accelerometer data; optimizing for on-device inference.

TimePilot **2026 - ongoing**

- Created an open-source Android library for time synchronization via SNTP. Available on Maven.

Lighthouse **2022 - ongoing**

- Created an open-source Android library for SSDP device discovery using Kotlin Coroutines. Available on Maven.